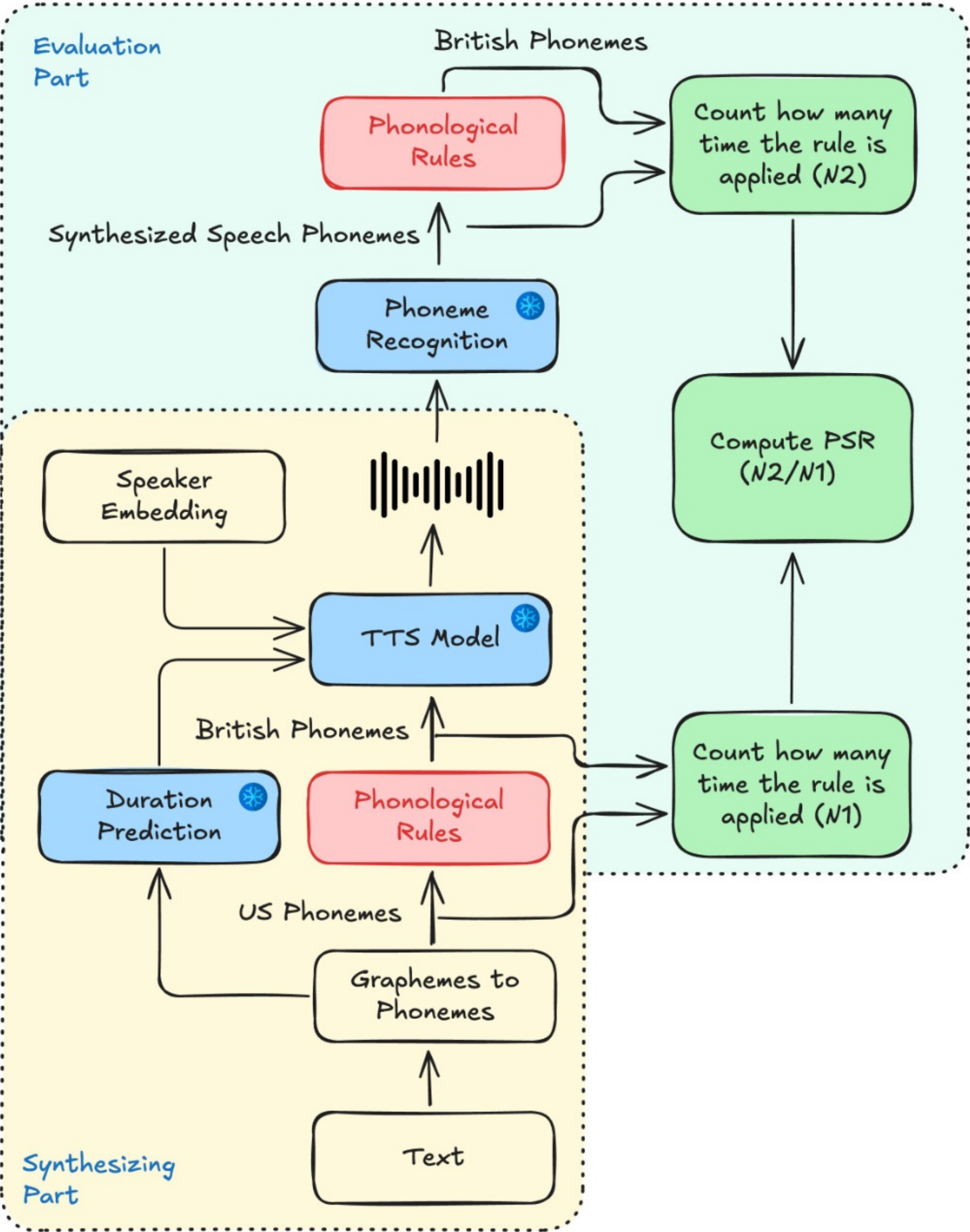


QUANTIFYING SPEAKER EMBEDDING PHONOLOGICAL RULE INTERACTIONS IN ACCENTED SPEECH SYNTHESIS

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Synthesis and Evaluation Pipeline



Phonological Rules

Flapping: Intervocalic /t/, [r] vs [t] (e.g., water: [wɑɹəʊ] vs [wɑtə])
Rhoticity: Post-vocalic /r/ is deleted or vocalized in British English (e.g. car: [kɑ] vs [kɑː])
Vowel Correspondences: TRAP, BATH, and GOAT (e.g., /bæθ/ vs /bɑθ/)

Effects of the Rule

- Total number of times we apply the rule (in thousands)
- N1 = applying on the phoneme input level
- N2 = applying on the phoneme output level
- $PSR = N2/N1$

	Flapping	Rhoticity	Vowel	All Rules
NA (N1)	12.8	83.5	125.1	221.4
NA (N2)	25.3	57.9	106.3	189.5
B w/o rules (N2)	12.3	57.4	101.7	171.5
B w/ rules (N2)	6.7	53.7	78.5	139.0

- We measure the effect of each rule by adding one or removing one from the whole set
- Same speaker embedding with different rules set.
- Evaluate with Vox-Profile, UTMOS, and PSR
- **Applying rule can push the output to be more accented**

Effects of Each Speaker Embeddings

- Different speaker embeddings give different level of control over the phonemes

	Accent Prob		Accent Sim		PSR ↓
	NA ↓	B ↑	NA ↓	B ↑	
Isabella	2.7	85.3	0.03	0.88	0.633
Isabella-R	1.1	90.1	-0.07	0.92	0.481 -15.2%
Lily	2.0	89.3	-0.01	0.91	0.660
Lily-R	0.9	91.6	-0.12	0.93	0.494 -16.6%
Fable	17.6	67.8	0.33	0.67	0.775
Fable-R	5.7	78.4	0.03	0.86	0.628 -14.7%
Daniel	4.7	89.8	0.07	0.86	0.706
Daniel-R	1.5	93.2	-0.07	0.93	0.543 -16.3%

	UTMOS	Accent Prob		Accent Sim		PSR ↓
		NA ↓	B ↑	NA ↓	B ↑	
NA Spk Emb	4.43	86.5	3.79	0.85	-0.05	0.856
+ All	4.42	58.8	17.3	0.74	0.21	0.827
B Spk Emb	3.74	17.6	67.8	0.33	0.67	0.775
+ Flapping	3.74	17.3	68.5	0.32	0.67	0.749
+ Rhoticity	3.73	9.5	68.8	0.14	0.78	0.739
+ Vowel	3.73	9.8	77.8	0.18	0.78	0.693
+ All	3.72	5.3	78.4	0.03	0.85	0.628
- Flapping	3.72	5.3	78.1	0.03	0.85	0.646
- Rhoticity	3.73	9.5	78.1	0.18	0.78	0.670
- Vowel	3.73	9.4	69.4	0.14	0.79	0.716

Kernel Density Estimation (KDE) of histograms on the number of changes of the rule

- Skewness toward smaller number of changes indicates how accent the output is.

